

ECONOMIC FORECAST

UK Hydrogen Market

2026-2031

Import and export outlook, industrial demand, energy use and market economics

Prepared for	Hydrogrid PLC
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Forecast horizon	2026-2031
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KEY JUDGEMENT UK hydrogen demand should expand only gradually through 2031 in the central case. The near-term market remains dominated by refinery and chemical uses; the main investable growth is likely to be low-carbon supply for selected industrial processes, hydrogen-to-power and fuel synthesis, clustered around infrastructure. Major commodity imports or exports are not yet the base case.

Treasury-style analytical format; not an HM Treasury publication or official government forecast.

Basis: Hydrogrid-provided "Hydrogen - Executive Summary" [1], supplemented by current DESNZ, parliamentary and UK government sector-plan publications available by 10 October 2026.

Executive summary

This note presents a scenario-based economic outlook rather than a point forecast. It takes the supplied executive summary as the initial technical frame, but updates the measured starting point using DESNZ's June 2026 hydrogen statistics. Those statistics put UK hydrogen demand at 12 TWh in 2025 and production at about 13 TWh; demand fell 8% and production fell 9% year-on-year. More than 80% of production was by-product hydrogen and refineries accounted for 75% of demand. The published data are "Official Statistics in Development", so the figures remain subject to methodological uncertainty. [1][2]

CENTRAL FORECAST FOR 2031 Hydrogen demand of about 13 TWh/year; low-carbon domestic production of around 1.8 TWh/year; direct molecular hydrogen imports and exports remain marginal. Imports of ammonia for cracking could create a small, location-specific hydrogen-equivalent import flow if the necessary terminal, conversion plant and offtake arrangements are delivered.

Indicator	2025 actual / baseline	2031 central case	Interpretation
UK hydrogen demand	12 TWh	13 TWh	Only modest net growth; legacy demand softens as new uses emerge.
UK hydrogen production	~13 TWh	~12.8 TWh modelled	By-product and conventional production declines; low-carbon output offsets part of the fall.
Low-carbon H ₂ output	Negligible at system level; exact volume not separately established in cited stats	~1.8 TWh	Indicative model assumption; depends on project delivery and load factors.
Molecular H ₂ / cracked-ammonia imports	No material national import market evidenced in the sources reviewed	0-0.5 TWh H ₂ -equivalent	Contingent, not a committed flow forecast.
Hydrogen / H ₂ -derived energy exports	No material flow assumed	0-0.1 TWh H ₂ -equivalent	Domestic offtake likely has priority until surplus production and logistics are established.
Gross annual commodity-equivalent value	Not a transparent, liquid national market	~£0.6-1.3bn	Illustrative cost-equivalent range, not reported revenue or GVA.

Main conclusions

- Hydrogen is still chiefly an industrial feedstock and refinery input, not a mass-market fuel. Refinery closures can shrink incumbent supply and demand together, complicating interpretation of market growth. [2]
- The government's 10 GW ambition and published end-use ranges are strategic pathways, not committed volumes. HAR2 project awards and the supporting networks are the key near-term delivery gates. [3][5][6]
- By 2031, trade is more likely to be selective imports of ammonia or other derivatives, and exports of equipment, engineering and services, than large-scale exports of hydrogen molecules. [3][7][8]
- For Hydrogrid, capital should be staged around credible offtake, a defined hydrogen standard and access to a cluster/network; avoid a merchant-trading thesis that assumes a liquid national spot market appears before 2031.

1. Starting point and forecast framework

Observed market in 2025

Measure	Latest measured position	Implication for the forecast
Production	Around 13 TWh in 2025; down 9% from 2024.	Industrial activity and refinery configuration remain key drivers.
Demand	12 TWh in 2025; down 8% from 2024.	Current market is not yet on a strong growth trajectory.
Production structure	More than 80% by-product; over 70% via catalytic reformers; 13% via steam methane reformers.	The existing supply base is not equivalent to a standalone low-carbon hydrogen industry.
Demand structure	Refineries: 75%; chemicals and non-energy use: 23%; reported other end-use sectors each below 1%.	New sectors are starting from a very small base.
Measurement caveat	DESNZ classifies the new series as Official Statistics in Development.	Use as a directional baseline; on-site production/consumption and missing suppliers may affect reconciliation.

The supplied report's broader framing remains useful: hydrogen can be produced through electrolytic ("green" or nuclear-powered "pink") routes, or through gas reforming with carbon capture ("blue"); long-distance imports may be carried as ammonia, liquid hydrogen or liquid organic hydrogen carriers; and domestic pipelines and storage become increasingly important as hubs scale. [1] This forecast does not treat every technically possible use as commercially likely by 2031.

Forecast principles

1. Separate the incumbent market from the emerging market. Existing hydrogen used in refining and chemicals is forecast independently from new demand in industrial heat, power, heavy transport and synthetic fuels.
2. Distinguish capacity from output. A project shortlist or announced gigawatt target is not equivalent to commissioned plant, contracted offtake or annual production.
3. Forecast traded molecules separately from hydrogen-derived products and technology exports. Imported ammonia used as a carrier and cracked to hydrogen is different from ammonia imported for fertiliser or chemical use; equipment exports are not hydrogen fuel exports.
4. Use scenarios rather than false precision. The central case discounts aspirational demand ranges for delivery risk, and the upside case assumes stronger policy, funding and coordinated infrastructure delivery.

POLICY SIGNAL Government strategy has described up to 10 GW of low-carbon production capacity by 2030, with a modelled output around 60 TWh/year once fully operational. The same roadmap says demand could reach up to 40 TWh/year in 2030 if sector demand is at the top of forecast ranges. These are high-end pathway estimates and substantially exceed this note's central 2031 forecast. [3]

2. Central economic forecast, 2026-2031

The central case assumes that some early electrolytic capacity is commissioned, project delivery is delayed relative to earlier timetables, and demand grows mainly in tightly defined industrial clusters. It assumes no broad hydrogen-heating conversion and no material national market for pure hydrogen imports or exports before 2031. Figures below are Hydrogrid analytical estimates, not official DESNZ forecasts.

Year	Demand (TWh)	Domestic production (TWh)	Low-carbon output (TWh)	Implied demand less production (TWh)
2025 actual	12.0	13.0	Not separately measured	-1.0
2026	11.8	11.9	0.1	-0.1
2027	11.8	11.8	0.25	0.0
2028	12.0	12.0	0.4	0.0
2029	12.2	12.2	0.7	0.0
2030	12.6	12.6	1.1	0.0
2031	13.0	12.8	1.8	0.2

Interpretation: the last column is an arithmetic difference between modelled aggregate production and demand, not a national trade forecast. Existing statistics do not yet provide a fully transparent, liquid market balance; hydrogen is often produced and consumed at the same facility. Negative values in the 2025 baseline reflect the reported 13 TWh production versus 12 TWh demand and should not be read as export volumes. [2]

Illustrative UK hydrogen demand paths

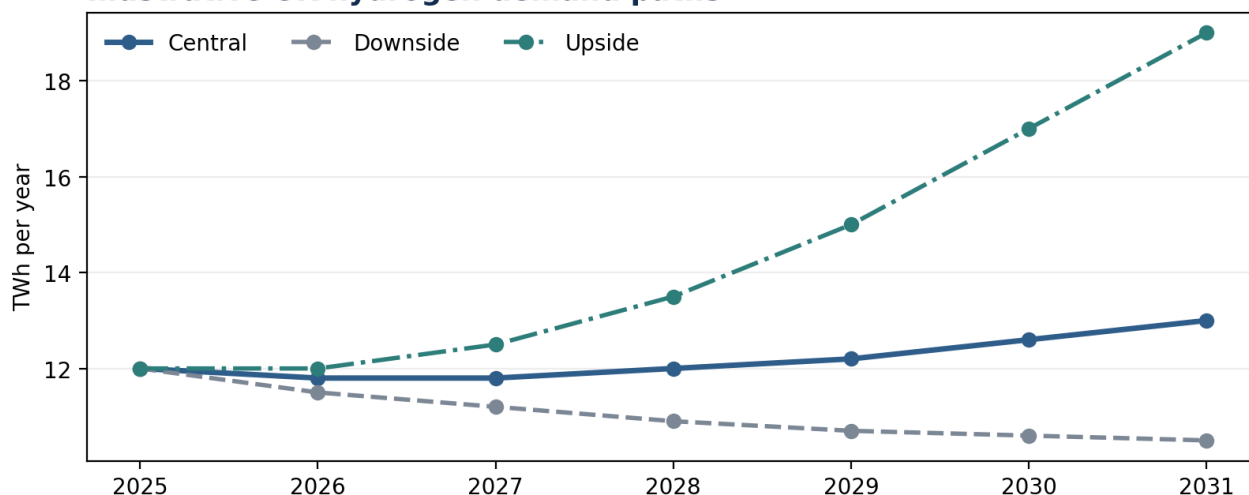


Chart 1. Scenario trajectories are illustrative. Central demand is 13 TWh in 2031; downside is 10.5 TWh and upside is 19 TWh. The upside remains below the government's 40 TWh high-demand pathway for 2030 because it assumes only partial materialisation of industrial, power and transport demand. [3]

Interpretation

- The incumbent market is forecast to decline slightly over time as refinery activity changes, while the new market compensates for part of that contraction.
- The central case does not require mass imports: it assumes most hydrogen continues to be produced domestically or at the point of use, with small import gaps possible in regional hubs.
- The key change through 2031 is market composition, not a tenfold increase in national volume. The 10x ambition is a longer-run, 2030s proposition in this forecast framework.

3. Hydrogen energy use and sector demand

Hydrogen demand is not synonymous with final energy consumption. In 2025, most refinery hydrogen was used in transformation processes, incorporating hydrogen into fuels rather than burning it directly. Chemicals and non-energy applications also remain important. [2] Policy and investment should therefore distinguish feedstock demand, heat demand and hydrogen used to make another energy carrier.

Sector / use	2025 measured picture	2031 central estimate	Assessment
Petroleum refining	75% of 12 TWh = ~9.0 TWh	7.5 TWh	Large incumbent market; sensitive to refinery closures, fuel mix and process configuration.
Chemicals and non-energy use	23% combined = ~2.8 TWh	2.6 TWh	Relatively sticky feedstock demand; low-carbon substitution is a near-term route to emissions reduction.
Industrial heat and steel / glass / ceramics	Each non-core sector currently very small	0.9 TWh	Potential industrial cluster demand, but conversion capital and fuel-price premiums are constraints.
Hydrogen-to-power / balancing	Below 1% of demand in the 2025 data	0.7 TWh	Strategically useful for dispatchable power and long-duration storage; fuel supply and business model remain gating issues.
Transport: HGVs, buses and refuelling	Below 1% in each reported other sector	0.4 TWh	Likely to remain concentrated in selected fleets, depots and corridors.
Hydrogen input to e-fuels / SAF and other synthesis	Not material in the 2025 demand profile	0.8 TWh	Potentially important longer-run feedstock; depends on project investment and product offtake.
Buildings / heat pilots	Below 1% in each reported other sector	0.1 TWh	Central forecast assumes no mass-market hydrogen heating or wide gas-grid conversion.
Total	12.0 TWh actual	13.0 TWh	Illustrative sector allocation, not official sector forecast.

Central 2031 demand by sector (illustrative)

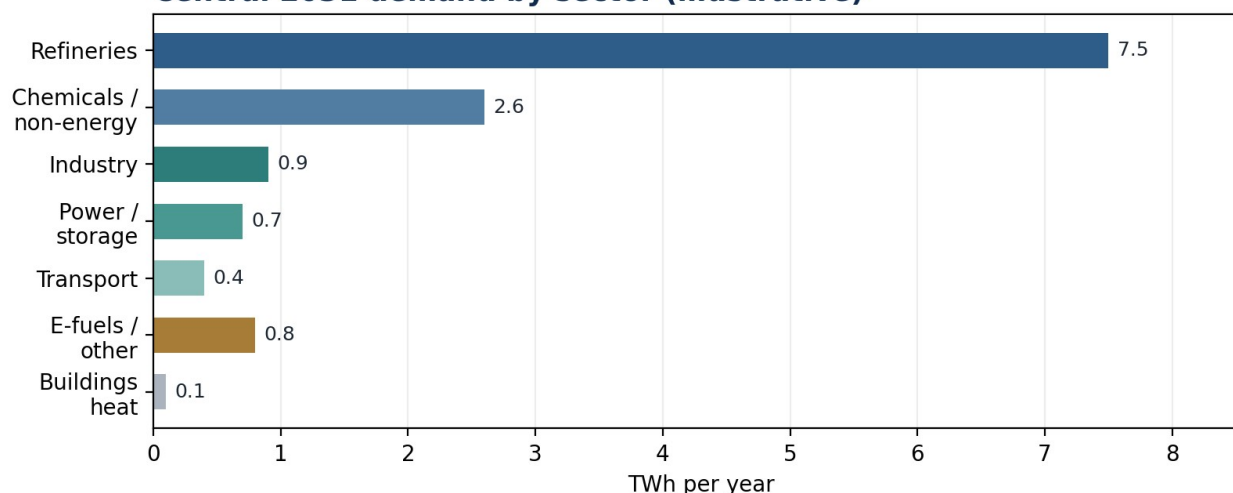


Chart 2. Central sector allocation in 2031 is an analytical distribution of the 13 TWh total. It is intended to show relative scale, not to imply a committed project pipeline or official sector-specific forecast.

DEMAND PRIORITY The strongest near-term case is replacement of existing high-carbon hydrogen in refineries and chemicals, followed by high-temperature industrial applications and selected hydrogen-to-power / synthetic-fuel projects. Domestic heating should not be the central investment case absent a major policy change. Government demand ranges span wide outcomes for industry, power, heat and transport. [3]

4. Import and export outlook

4.1 Direct hydrogen trade

A material UK market for cross-border pure hydrogen is unlikely to develop quickly by 2031. Hydrogen remains difficult to transport economically at long distances, the national pipeline/storage system is still being developed, and the first regional transport-and-storage network is expected to become operational from 2031. [6] This supports a central case of small, project-specific import and export flows rather than a liquid, nationally priced import-export market.

Trade channel	Central case by 2031	Upside range	What must happen
Pure molecular hydrogen imports	Negligible; no major pipeline trade assumed	0-0.5 TWh/year	Contracted supply, compatible port/pipeline facilities, certification and a buyer willing to pay delivered cost.
Ammonia imports cracked to hydrogen	0-0.5 TWh/year H ₂ -equivalent; point estimate 0.2	1-3 TWh/year H ₂ -equivalent	Import terminal, cracking plant, handling approvals, reliable shipping, conversion economics and offtake.
Hydrogen exports / exports in energy carriers	0-0.1 TWh/year H ₂ -equivalent	0.5-1.5 TWh/year H ₂ -equivalent	Domestic production surplus, export certification, port / conversion infrastructure and a bankable overseas buyer.
Imports of hydrogen-derived fuels / feedstocks	Selective, but not quantified as pure H ₂ trade	Material in some products, hard to aggregate	Trade depends on product-specific markets such as ammonia, methanol and sustainable aviation fuel.
Hydrogen equipment / engineering exports	Not expressed as TWh	Government estimates £0.8-2.2bn export opportunity through 2030 for UK hydrogen production equipment and fuel cells	Manufacturing scale, order wins, supply-chain competitiveness and export finance.

The import terminal at Immingham is an important potential route: development consent was granted in February 2025 for a liquid-bulk terminal and associated facility intended to store imported ammonia and process it into hydrogen. Consent is not proof of final investment decision, commissioning or a committed flow of hydrogen. [8] The central estimate therefore treats it as a potential option, not as assured supply.

4.2 Why ammonia matters more than liquid hydrogen in the near term

- Ammonia is already a traded chemical and can be transported in bulk. The conversion step back to hydrogen carries an efficiency and cost penalty, and ammonia is toxic, so safety and offtake design matter. [1]
- Liquid hydrogen requires energy-intensive liquefaction and cryogenic handling; long-distance transport can consume a material share of the delivered energy. [1]
- Pipelines become attractive for regional, high-volume flows but depend on network coordination, storage and anchor customers. The expected first network from 2031 points to a hub-based market rather than a national spot market in 2026-2030. [6]

4.3 Exports: fuel or technology?

For UK companies, the more credible near-term export opportunity may be equipment, fuel cells, electrolyser systems, project engineering and CCUS-related services. The government Clean Energy Industries Sector Plan states that the export market for UK manufacturers of CCUS-enabled and electrolytic hydrogen production equipment and fuel cells could be £0.8-2.2 billion in the period to 2030, potentially £5.8-9.8 billion by 2050. These are estimated market opportunities, not realised exports or a guaranteed Hydrogrid revenue line. [7]

5. Market economics and scenarios

5.1 Indicative commodity value

There is no fully transparent UK merchant hydrogen price series in the evidence reviewed. Much of the existing market is captive production/consumption, while early low-carbon projects rely on revenue support. As a reference point, the 11 HAR1 electrolytic projects were agreed at a weighted-average strike price of £241/MWh, expressed by government as £175/MWh in 2012 prices. A support strike price is not the same as an unassisted spot-market price, but it illustrates the cost gap for early low-carbon supply. [4]

2031 central cost proxy	Volume assumption	Indicative unit-cost proxy	Implied annual value
Existing / conventional hydrogen	11.2 TWh	£30-80/MWh	£0.34-0.90bn
Low-carbon hydrogen	1.8 TWh	£120-240/MWh	£0.22-0.43bn
Total commodity-equivalent	13.0 TWh	Weighted mixed basket	~£0.6-1.3bn/year

The cost bands above are explicit modelling assumptions for sensitivity analysis, not official market prices. The total is a procurement-cost-equivalent estimate; it excludes conversion losses, transport and storage tariffs, network capital, subsidy transfer payments, tax, and downstream value added. Avoid treating it as GDP, sales revenue or a bankable addressable market without project-specific data.

5.2 Three scenarios for 2031

Variable	Downside / delayed	Central / staged	Upside / accelerated
Total hydrogen demand	10.5 TWh	13 TWh	19 TWh
Low-carbon domestic output	0.6 TWh	1.8 TWh	4.0 TWh
H ₂ -equivalent imports	0-0.2 TWh	0-0.5 TWh	1-3 TWh
H ₂ / derived-energy exports	Near zero	0-0.1 TWh	0.5-1.5 TWh
Demand pattern	Incumbent refining/chemical base declines; new projects delayed.	Incremental industrial, power and fuel-synthesis demand partially offsets incumbent decline.	More industrial conversions, power demand and e-fuel projects proceed with stronger policy and funding.
Conditions	HAR2 slippage, weak offtake, power-cost disadvantage, limited infrastructure.	Some HAR1/HAR2 delivery; early hubs; no broad heating conversion.	More awards, faster project execution, anchor offtakers, available renewable power and working trade logistics.

5.3 Investment and public-finance implications

- Production support remains a key price-risk bridge. HAR1 has committed substantial long-term revenue support for 125 MW of projects; HAR2 aims to select from a 27-project shortlist but award timing has slipped relative to earlier published expectations. [4][5]
- Transport and storage are enabling assets, not optional extras. The government has announced over £500m of support for hydrogen infrastructure following the 2025 Spending Review, and expects the first regional network from 2031. [6]
- Investment appraisal should consider whole-chain cost: clean power, electrolyser utilisation, hydrogen conditioning, storage, delivery, conversion at the user and any subsidy. Low equipment CAPEX alone does not establish a competitive delivered hydrogen price.

6. Implications for Hydrogrid PLC

The forecast supports an option-based strategy: secure a position in selected regional hydrogen value chains without underwriting national-scale demand or export volumes before policy support and infrastructure become operational.

Priority	Recommended response	Decision trigger / evidence
1. Anchor demand first	Target industrial and chemical users with a documented emissions-reduction need and a clear buyer credit profile.	Signed offtake contract, defined annual volume, acceptable delivered-price formula and low-carbon standard.
2. Concentrate on hubs	Screen locations near industrial demand, low-carbon power, CCUS/storage where relevant, salt-cavern options and the first regional network.	Confirmed network route and milestones; credible storage/transport business model and permit pathway.
3. Keep trade optionality	Evaluate ammonia-cracking or derivative-import routes as contingent balancing supply, not as the core central forecast.	Terminal and conversion FID, contracted source of ammonia, safety approvals, conversion costs and offtake.
4. Do not conflate export markets	Separate export plans for physical H ₂ / carriers from sales of equipment, engineering, control systems and services.	Product-specific export pipeline, verified customer demand, certification and finance.
5. Stage capital	Release development capital at defined policy and commercial gates rather than against headline GW ambitions.	HAR outcome, permit, grid connection, power supply, offtake, EPC price and commissioning plan.
6. Build monitoring discipline	Maintain a quarterly dashboard of project awards, FIDs, commissioning, output, demand contracts and import-terminal milestones.	Use published capacity, actual output and contracted demand separately; record source date and confidence.

Risk register

Risk	Likely consequence	Monitor
Policy / award delay	Production capacity and offtake slip beyond forecast.	HAR2 awards, HAR3 launch, support contract signature dates.
Power price and grid constraints	Electrolyser utilisation or economics weaken.	Electricity connection dates, renewable contracts, curtailment and load factors.
Weak end-user conversion	Low-carbon supply lacks buyers; subsidy per tonne rises.	Binding offtake, conversion CAPEX, industrial operating margins.
Infrastructure sequencing	Producers and users remain disconnected or lack storage.	HTBM/HSBM allocations, route selection, storage FID and 2031 network delivery.
Trade / carrier economics	Ammonia cracking or liquefaction increases delivered cost.	Shipping assumptions, carrier losses, cracking CAPEX/OPEX and emissions standard.
Legacy market contraction	Refinery closures reduce established hydrogen supply and demand.	Refinery operating rates/closures, chemical-site plans, annual DESNZ demand series.

STRATEGIC CONCLUSION For 2026-2031, Hydrogrid should underwrite a regional industrial hydrogen market that grows steadily but remains small, with traded H₂ volumes limited and project economics heavily influenced by power price, revenue support and the timing of networks. Treat export technology and engineering as a distinct upside option; treat bulk H₂ exports and imports as conditional on surplus output and delivered-cost economics.

7. Assumptions, definitions and limitations

Forecast assumptions

- Central demand moves from the measured 12 TWh in 2025 to 13 TWh in 2031. This reflects falling or flat legacy demand offset by only partial adoption in new sectors.
- Central low-carbon domestic output increases to 1.8 TWh in 2031. This assumes staged, delayed partial commissioning of supported projects; it is not a published capacity projection.
- The upside case assumes stronger public support, accelerated HAR allocation and execution, credible industrial offtake, renewable electricity access, and operating transport/storage arrangements.
- No mass-market hydrogen heating or broad gas-grid conversion is included in the central case. The upside case allows for limited additional demand where policy and local economics support it.
- Trade estimates are expressed as annual hydrogen-equivalent energy in TWh. Imports of ammonia for uses other than cracking to hydrogen are not counted as hydrogen imports. Export estimates for hydrogen-derived products are not official trade statistics.
- The economic value estimate uses modelling unit-cost bands and is intended for scenario comparison only. It should not be read as actual aggregate market turnover, GVA, tax receipts or an investment valuation.

Key definitions

Term	Definition used in this note
Hydrogen demand	Hydrogen consumed as measured or estimated in the national supply/demand series, including feedstock and transformation uses, not just hydrogen burned as fuel.
Low-carbon hydrogen	Hydrogen produced through electrolysis using qualifying low-carbon electricity or through reforming with carbon capture, subject to applicable UK standards.
H ₂ -equivalent trade	Energy content of hydrogen imported/exported directly or recovered from a carrier such as ammonia. Excludes ammonia trade for unrelated chemical use.
Domestic production	Includes existing by-product / conventional hydrogen plus modelled low-carbon production. Actual national data do not provide a fully reconciled merchant-market balance.
Strike price	Contract benchmark used in revenue support; not necessarily the price paid by an end-user or a merchant market-clearing price.

Sources and references

[1] Hydrogrid-provided source report, "Hydrogen - Executive Summary" (6 pages), user-uploaded file. Used for production pathways, transport/carrier options, storage and sector-use framing.

[2] Department for Energy Security and Net Zero (DESNZ), "Hydrogen production and demand in the UK, 2022 to 2025", Energy Trends, published 30 June 2026 and updated 6 July 2026. <https://www.gov.uk/government/publications/energy-trends-june-2026-special-feature-article-hydrogen-production-and-demand-in-the-uk-2022-to-2025>

[3] DESNZ, "Hydrogen production delivery roadmap", published 14 December 2023. Includes production ambition, modelled 60 TWh output from up to 10 GW and 2030 sector demand ranges. <https://www.gov.uk/government/publications/hydrogen-production-delivery-roadmap/hydrogen-production-delivery-roadmap>

[4] DESNZ, "Hydrogen Production Business Model / Net Zero Hydrogen Fund: HAR1 successful projects", updated 14 December 2023. 11 projects totalling 125 MW; weighted-average strike price £241/MWh (£175/MWh in 2012 prices). <https://www.gov.uk/government/publications/hydrogen-production-business-model-net-zero-hydrogen-fund-har1-successful-projects>

[5] UK Parliament, written answer on the status of HAR2, answered 15 June 2026. DESNZ stated due diligence and cost assurance on shortlisted projects had finished, with the invite-to-offer stage still pending. <https://questions-statements.parliament.uk/written-questions/detail/2026-06-01/HL439/>

[6] DESNZ, "Clean flexibility roadmap", updated 13 July 2026. Describes over £500m support for hydrogen infrastructure and the expected first regional transport/storage network from 2031. <https://www.gov.uk/government/publications/clean-energy-industries-sector-plan/clean-energy-industries-sector-plan>

[7] UK Government, "Clean Energy Industries Sector Plan", 2025. Hydrogen equipment and fuel-cell export opportunity of £0.8-2.2bn to 2030, potentially £5.8-9.8bn by 2050. <https://www.gov.uk/government/publications/clean-energy-industries-sector-plan/clean-energy-industries-sector-plan>

[8] Planning Inspectorate / UK Government, "Immingham Green Energy Terminal development consent decision announced", 6 February 2025. The consented project includes ammonia import, storage and processing into hydrogen. <https://www.gov.uk/government/news/immingham-green-energy-terminal-development-consent-decision-announced>

Prepared 10 October 2026. Public sources checked for this note were those available by the report date. The forecast is Hydrogrid-specific analytical work and is not endorsed by HM Treasury, DESNZ or any other public authority.